SafeX Solutions AI/ML Internship · Week 5 Progress Report

Project Title: Insurance & Finance Lead Qualifier | **Developer:** Ali Zaib | **Group:** 54 | **Status:** Completed & Submitted

1. Executive Summary

Insurance and wealth-management sales teams receive a high volume of leads of wildly varying quality — some ready to buy immediately, others years away from a decision, and many with budget or risk profiles that make them a poor fit for the requested coverage. During Week 5, I built an Insurance & Finance Lead Qualifier that scores incoming leads on five weighted, explainable factors and sorts them into clear sales/underwriting priority tiers, so a sales rep or underwriter can immediately see which leads to act on first — and why.

2. Problem Statement & Business Opportunity

- **The Problem:** Sales reps and underwriters manually review every incoming lead with no consistent prioritization, wasting time on low-quality or poor-fit leads while hot leads sit in the same queue.
- **Target Audience:** Insurance agencies and wealth-management firms processing policy applications across corporate, fleet, and individual product lines.
- **Value Proposition:** Instantly ranks a batch of leads by qualification score, flags specific risk/budget issues for manual review, and gives a transparent breakdown a sales rep can act on immediately.

3. Technical Approach & Engineering

Why a rubric, not a black-box ML model?

For an underwriting-adjacent use case, transparency matters more than squeezing out marginal accuracy — a sales rep or underwriter needs to see why a lead scored the way it did. A weighted, explainable rubric can show its work; a black-box classifier can't.

Scoring rubric (0-100 total)

Factor	Points	What it measures
Policy Value Fit	0-25	Value/complexity of the requested product
Coverage/Budget Fit	0-20	Does stated budget realistically support requested coverage?
Urgency/Timeline	0-20	How soon the applicant wants to decide
Risk Profile	0-20	Prior claims, age band, occupation risk
Engagement/Channel	0-15	Referral/broker channels convert better than cold outreach

4. Verification & Testing Results

- **Unit tests:** 11 pytest functions covering scoring correctness, tier boundaries, all five validation error paths, and batch-scoring behavior. All 11 pass.
- **Sample dataset validation:** ran the full 40-lead synthetic sample batch through the scorer — produced a realistic tier distribution with no crashes or nonsensical scores.
- **End-to-end UI testing:** verified via Streamlit's AppTest — submitted the single-lead form and ran batch scoring against the live rendered UI, confirming no runtime exceptions.
- **Standalone deployment package:** verified independently, confirming the standalone app produces identical results to the in-suite version. Deployed live to Streamlit Community Cloud.

5. Problems Encountered & Solutions

- **Problem:** the original module stub scored leads with only two branches, which couldn't meaningfully differentiate leads or explain its reasoning. **Solution:** redesigned as a five-factor weighted rubric with per-factor point breakdowns.
- **Problem:** a coverage amount alone doesn't indicate lead quality. **Solution:** added a coverage/budget-fit check against an industry rule-of-thumb premium rate, surfacing mismatches as specific, actionable flags.
- **Problem:** the batch dashboard initially listed leads in a plain table with no at-a-glance prioritization. **Solution:** added explicit red/yellow/green row-level color coding by tier.

6. Next Week's Plan (Week 6)

Package this tool as a sellable commercial service for insurance agencies/brokers: tiered pricing, an ROI calculator for discovery calls, and outreach materials — then conduct genuine outreach to real prospective agencies as required by the Week 6 business-development exercise.